problems across several complexity classes, these limitations can be alleviated with additional memory (e.g., scratchpads) (Liu et al., 2024). However, this still requires generating vast amounts of tokens to solve a problem (Peng et al., 2024; OpenAI, 2024). While these works provide insights into the theoretical computational complexity of transformers, in practice, it remains unclear whether these LLMs can perform formal logical reasoning to solve tasks.

There is a considerable body of work suggesting that the reasoning process in LLMs is not **formal** (Kambhampati, 2024; Valmeekam et al., 2022, 2024), even though it appears that these models understand symbols and can work with them to some limited degree (Boix-Adserà et al., 2024). Instead, LLMs likely perform a form of probabilistic **pattern-matching** and searching to find closest seen data during training without proper understanding of concepts. While this process goes beyond naive memorization of words and the models are capable of searching and matching more abstract reasoning steps, it still falls short of true formal reasoning. For instance, Jiang et al. (2024) show, with statistical guarantees, that most LLMs still struggle with logical reasoning due to strong token bias, where the reasoning output of the model changes when a single token of input changes. This aligns with our results, which indicate that the performance of models on different instances of the same mathematical question can vary greatly from one instance to another. Li et al. (2024b) prove that a single transformer layer learns a one-nearest neighbor, which could explain why the reasoning of models is highly sensitive to input tokens. Schaeffer et al. (2023) argue that when a task requires emitting multiple tokens correctly, the probability of answering correctly decreases exponentially with the number of tokens. Dziri et al. (2023) represent reasoning tasks as computation graphs and find that full computation subgraphs appear much more frequently in training data for correct predictions than incorrect ones. Razeghi et al. (2022) show a correlation between frequency in training and test performance, supporting the pattern matching hypothesis.

Our work builds upon these findings by introducing `GSM-Symbolic`, an improved benchmark using symbolic templates to generate diverse question variants. This allows us to study mathematical reasoning ability beyond a single performance metric. By evaluating performance on different instantiations and difficulty levels, we draw a comprehensive picture of LLMs' reasoning capabilities. Our findings support the hypothesis that current LLMs are not capable of performing formal mathematical reasoning and pave the way for further research on this important topic.

3 GSM-Symbolic

The `GSM8K` dataset (Cobbe et al., 2021) includes over 8000 grade school math questions and answers, divided into 7473 training and 1319 test examples. As shown in Fig. 1, the questions are relatively simple, requiring knowledge of only the four main arithmetic operations. However, since `GSM8K` is a single, popular test set, there is a risk of data contamination, and performance may change significantly with minor modifications to the questions. These limitations have led to efforts to generate new datasets and variants. `iGSM` (Ye et al., 2024) is a math dataset created through a synthetic pipeline that captures parameter dependencies in a hierarchical and graph structure. `GSM-IC` (Shi et al., 2023) shows that irrelevant context can impair LLM performance, focusing on prompting techniques. Our work, however, suggests a more fundamental issue: LLMs struggle even when given multiple shots of the same question, indicating deeper challenges in problem-solving that cannot be resolved with few-shot prompting or fine-tuning on unseen distractions or variations of the same or different difficulty levels. `GSM-Plus` (Li et al., 2024a) introduces variants of GSM8K questions but lacks symbolic templates and has a fixed size and difficulty. `GSM1K` (Zhang et al., 2024) mirrors the style and complexity of `GSM8K` to identify systematic overfitting in existing models, but has a fixed number of examples, and is not publicly available for researchers.

While the mentioned benchmarks offer a single performance metric on a fixed number of questions, we argue that viewing LLM performance as a distribution across various problem instances provides deeper insights. The design of `GSM-Symbolic` enables the generation of numerous instances and allows for finer control over question difficulty. We believe our paper contributes to this direction by offering a reliable evaluation framework that underscores the importance of generating multiple instances to assess LLMs' mathematical capabilities and their robustness to diverse problem difficulties and augmentations.

3.1 GSM-Symbolic: Template Generation

Given a specific example from the test set of `GSM8K`, we create parsable templates as shown in Fig. 1 (right). The annotation process involves identifying variables, their domains, and necessary conditions to ensure the correctness of both the question and the answer. For instance, since the questions are grade-school level, a common condition is divisibility to ensure the answer is a whole number. We use common proper names (e.g., persons, foods, currencies) to streamline template creation. After creating the templates, we apply several automated checks to ensure the annotation process is correct. For example, we verify that none of the original variable values appear in the template. We also check that the original values satisfy all conditions and that the final answer matches the original question's answer. Once data are generated, 10 random samples per template are reviewed manually. As a final automated check, after evaluating all models, we verify that at least two models answer each question correctly; otherwise, the question is reviewed manually again.

3.2 Experimental Setup

While we provide further details on our experimental setup and evaluation in the Appendix, we briefly review the important aspects here:
Models. Throughout this work, we report on more than 20 open models of various sizes, ranging from 2B to 27B. Additionally, we include state-of-the-art closed models such as GPT-4o-mini, GPT-4o, o1-mini, and o1-preview. To conserve space, we present results for a few selected models in each experiment, but the full results for all models are available in Tab. 1 of the Appendix A.2.
Evaluation Setup Overall, for this work, we conducted nearly 500 total evaluations on various setups. To this end, we maintained a manageable dataset size by using 100 templates and generating 50 samples per template, resulting in 5000 total examples for each benchmark. Therefore, we have 50 datasets of 100 examples each, where each example is a mutation of one of the original 100 examples from `GSM8K`. Unless stated otherwise, we follow a common evaluation setup on GSM8K and other math benchmarks that includes Chain-of-Thought (CoT) prompting with 8-shots with greedy decoding. However, we note that in our preliminary experiments, the number of shots did not significantly change the performance and conclusions. We provide our prompt template in Fig. 9.

4 Experiments & Results

In this section, we present our main results and postpone complementary findings to the Appendix. We begin our experiments by addressing an important question regarding the reliability of current reported metrics on `GSM8K`. By studying the *distribution* of performance on `GSM-Symbolic`, we demonstrate notable performance variation. More importantly, we observe that the performance of models drops on `GSM-Symbolic` (Sec. 4.1).
Next, we investigate the fragility of reasoning in LLMs by comparing performance distributions when only proper names are changed versus when values and numbers are altered. Our findings indicate

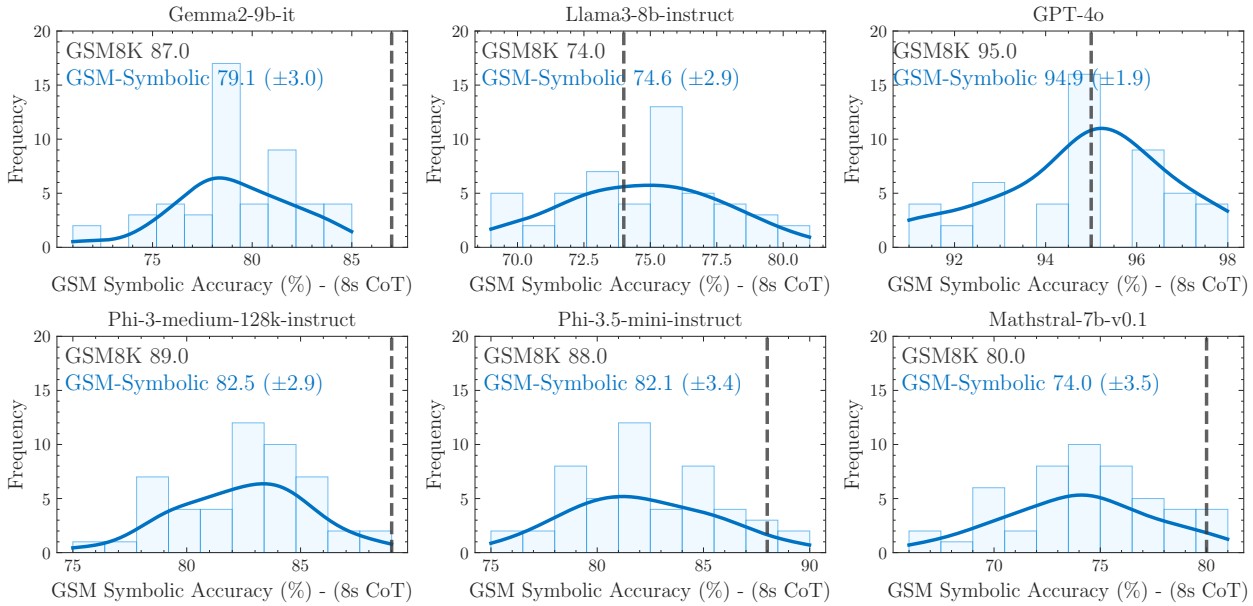

Figure 2: The distribution of 8-shot Chain-of-Thought (CoT) performance across 50 sets generated from `GSM-Symbolic` templates shows significant variability in accuracy among all state-of-the-art models. Furthermore, for most models, the average performance on `GSM-Symbolic` is lower than on `GSM8K` (indicated by the dashed line). Interestingly, the performance of `GSM8K` falls on the right side of the distribution, which, statistically speaking, should have a very low likelihood, given that `GSM8K` is basically a single draw from `GSM-Symbolic`.

that the original `GSM8K` performance of models is much closer to the performance distribution when only names are changed. However, performance drops more significantly when values are changed, with this trend continuing as both changes are applied simultaneously (Sec. 4.2). We then examine the impact of question difficulty, as indicated by the number of clauses added to or removed from the questions. Our results show that as the number of clauses increases, average performance drops, and the variance in performance increases consistently across all models (Sec. 4.3).
Finally, in Sec. 4.4, we tackle a more fundamental question: whether the models truly understand the mathematical concepts. We show that, likely due to potential pattern matching and the fact that the training distribution of models included only necessary information for solving questions, adding seemingly relevant clauses to the question that do not impact the reasoning process required to solve it significantly drops the performance of all models.

4.1 How Reliable Are the Current GSM8K Results?

As our first experiment, we evaluate the performance of several state-of-the-art models on `GSM-Symbolic`. The number of samples and difficulty can be adjusted by modifying variable domains, as we will see in subsequent sections. Fig. 2 shows the empirical distribution of the performance of models on `GSM-Symbolic` computed on these 50 datasets. As shown, all models exhibit a non-negligible variance across different sets. For instance, for the Gemma2-9B, the gap between the worst performance and the best performance is more than 12%, while for Phi-3.5-mini, this gap is around 15%. It is interesting that this variation even exists, as the only differences across different instances of each question are the changes in names and values, while the overall reasoning steps needed to solve a question remain the same.